Medical School Environments and Physician Treatment Styles

SUPPLEMENTAL APPENDIX

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A Sample Construction Details

This section provides additional detail on the construction of our cardiologist-year analytical panel, the medical school-to-HRR crosswalk, the AHA training-period cath lab measure, and the NIH ExPORTER ranking panel.

A.1 Cardiologist Assignment and Volume Restriction

For each NSTEMI episode, we assign a single attending cardiologist following the procedure of Molitor (2018). The hierarchy prioritizes the first cardiologist who delivers a consultation (CPT codes 99251–99255 prior to their elimination by CMS in January 2010, with the hierarchy collapsing to first cardiologist of record for inpatient care for episodes thereafter), the first cardiologist of record on the inpatient stay, and the first cardiologist seen by the patient in carrier-line claims within 90 days of the index admission. We exclude episodes for which no cardiologist could be assigned and patient-years for which Medicare fee-for-service coverage was not present for the entire year. Cardiologists are identified in MD-PPAS using primary-specialty fields for cardiology, interventional cardiology, clinical cardiac electrophysiology, and advanced heart failure and transplant cardiology.

We restrict to cardiologist-years with at least 11 NSTEMI patients. The threshold provides stable per-physician estimates of catheterization rates and aligns with the CMS cell-size

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threshold for public-use files. A trade-off of the volume restriction is that established, higher-volume cardiologists are over-represented in our analytical sample relative to the broader cardiologist population.

A.2 Residualized Catheterization Rate

Our outcome is the volume-weighted mean residualized cath-within-two-days indicator at the cardiologist-year level. At the patient level, we estimate a linear probability model regressing the cath-within-two-days indicator on the 31 Elixhauser comorbidity flags and year fixed effects, retaining the residual as the patient-level outcome. We then aggregate to the cardiologist-year by taking the volume-weighted mean residual. The residualization removes mechanical variation across cardiologists that would arise from differences in patient comorbidity mix.

Elixhauser comorbidities are constructed from a lookup table of approximately 2,600 ICD-9 and ICD-10 diagnosis codes mapped to 31 categories, drawn from the cardiologist's NSTEMI-episode and 12-month-look-back claims. A given diagnosis code may map to multiple Elixhauser categories, so the lookup is implemented as a join rather than a categorical recoding.

A.3 Medical School-to-HRR Crosswalk

The medical school HRR is constructed in three stages. First, we obtain each cardiologist's medical school name and graduation year from the CMS Physician Compare National Downloadable File. Second, we map medical school names to ZIP codes using a hand-curated crosswalk restricted to LCME-accredited U.S. allopathic medical schools, validated against the LCME accreditation directory. Of 187 medical schools in our crosswalk, 186 resolve via the LCME accreditation join, and one historically defunct school resolves via a separate fallback ZIP. Schools coded in Physician Compare as "OTHER" are excluded; the OTHER category captures most foreign-trained physicians and a small number of unmatched U.S. schools. Third, we map ZIP codes to hospital referral regions using the Dartmouth Atlas ZIP-to-HRR crosswalk (vintage 2015).

Our analytical panel restricts to allopathic MD graduates because only roughly 11% of U.S. medical school graduates are osteopathic (DO), with an even smaller share among cardiologists (AAMC, U.S. Physician Workforce Data Dashboard).

A.4 AHA Training-Period Cath Lab Measure

Cath lab availability is recorded in the AHA Annual Survey CCLABHOS variable from 1994 onward and in the predecessor CCLAB82 variable from 1980–1993. Both are recoded to a binary indicator of direct hospital provision of cardiac catheterization. The HRR-year cath lab share is then computed as the share of AHA-respondent hospitals in each HRR-year for which the binary indicator equals one. The aggregated AHA series exhibits a smooth secular trend across the 1994 questionnaire revision, with the national HRR-mean cath lab share rising from approximately 0.14 in 1980 to approximately 0.39 in 2003. We interpret the smooth trend as evidence that the recoding from CCLAB82 to CCLABHOS does not introduce a measurement break.

For each cardiologist, we match the AHA HRR-year cath lab share at the year the cardiologist began medical school, defined as graduation year minus three (the modal duration of allopathic medical school in the United States). AHA cath lab data are available 1980–2003, so the year-matching produces clean assignments for cardiologists with graduation years 1983 through 2006. We restrict our analytical sample to this graduation-year window.

A.5 Year-Matching and Cohort Variation

Our preferred specification identifies β_{train} off variation in cath lab availability across graduation cohorts who attended medical school in the same HRR. The variation comes from two sources. First, the rollout of cath labs occurred at different times across HRRs over the 1980–2003 period, so the same medical school HRR exposed different cohorts to different cath lab shares. Second, even within HRRs that had cath labs early, the share of hospitals offering cath services continued to rise across the rollout period. We rely on both sources of within-HRR temporal variation in our identification.

A.6 NIH ExPORTER Ranking Panel

To distinguish procedural exposure from medical school prestige, we construct a year-matched panel of medical school NIH research funding from the NIH ExPORTER bulk files. We aggregate annual project-level award totals (1985–2025) to the medical school level using the institution-name field, with consolidation of name variants (abbreviations, mergers, renamings) by hand. Each medical school then receives a within-year rank in total NIH funding. The constructed series is validated against the published Blue Ridge Institute for Medical Research rankings, with a Pearson correlation across overlapping years above 0.99.

For each cardiologist, we match the medical school NIH rank in the year of medical school start (graduation year minus three), parallel to the AHA cath lab matching procedure. We use both continuous $-\log(\text{rank})$ and tier indicators (Top 10, Top 11–25, Top 26–50, Top 51–100, with unranked schools as the reference) in our specifications.

A.7 GME Duration Filter

We drop physician-years for which the implied training pipeline is incompatible with the cardiologist's observed graduation year and subspecialty. The required GME duration after medical school is six years for general cardiology, seven years for interventional cardiology, eight years for clinical cardiac electrophysiology, and seven years for advanced heart failure. Physician-years implying a training pipeline shorter than the GME duration are dropped. The filter removes approximately 70 of approximately 31,000 physician-year observations.

B Cross-Sectional Balance, by Subspecialty

Table B.1 reports the training-period cath lab comparison between cross-sectional movers and stayers separately within each cardiology subspecialty. The ordering is similar across subspecialties, with movers in each subspecialty coming from medical school HRRs with lower cath lab share than stayers in that same subspecialty. The differences are statistically significant in general cardiology, interventional cardiology, and electrophysiology. Advanced heart failure has too few stayers (six cardiologists) to support a precise within-subspecialty test.

Table B.1. Cross-sectional balance by cardiology subspecialty

	Stayers		Movers			
Subspecialty	N	Train. cath	N	Train. cath	Diff.	p
General Cardiology	491	0.034	3,075	0.021	−0.013	<0.001
Interventional	136	0.042	875	0.027	−0.015	<0.001
Electrophysiology	23	0.041	173	0.021	−0.020	0.007
Adv. Heart Failure	6	0.022	26	0.023	0.001	0.974

C Head-to-Head with Current Peer Environment

The within-origin specification in the main paper absorbs the current peer environment through destination HRR fixed effects, which condition out all cross-HRR variation in peer cath rates. A natural alternative is to replace the destination HRR fixed effects with a continuous control for current peer cath culture, which allows training-period exposure and current peer environment to be compared on the same scale. Table C.2 reports the head-to-head specification, in which both the training-period cath lab share and the leave-one-out current-HRR peer cath rate (z-scored) enter as regressors, with year fixed effects and clustering at the medical school HRR level.

Table C.2. Head-to-head specification, training cath share and current peer cath culture

	Training only	Current only	Both
Training-HRR cath lab share (z)	0.006***		0.006***
	(0.002)		(0.002)
Current-HRR cath culture (z)		0.004	0.004
		(0.002)	(0.002)
Practice HRR FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Observations	10,729	10,729	10,729

The training-period cath lab share retains a statistically significant marginal effect of 0.006 (SE 0.002, $p < 0.001$), while the current-HRR peer cath culture coefficient is smaller (0.004, SE 0.002) and not statistically distinguishable from zero at the 5% level. A cardiologist who trained in a procedurally rich market continues to catheterize at higher rates than one who trained in a procedurally poor market, even when current peer environment is included as a continuous covariate rather than absorbed by HRR fixed effects.

D NIH Rank Specifications

The main paper documents that medical school NIH-funding rank does not predict later catheterization rates, in contrast to the cath lab availability measure, which does. This section reports additional rank specifications that test the robustness of the rank null result.

The continuous $-\log(\text{rank})$ specification, the binary Top-25 indicator, and the four-tier specification all yield small and statistically insignificant coefficients on the cardiologist-year panel. Restricting to the mover sample produces similar nulls. The interpretation is that whatever channel rank captures, it does not show up in NSTEMI cath rates. The rank-by-cath stratification reported in the main paper further indicates that the cath lab imprint is similar in magnitude within the Top-25 and Below-Top-25 subsamples, confirming that cath lab availability and rank capture nearly orthogonal dimensions of the medical school environment.

E Doximity Match Procedure

Section ?? uses Doximity profile data to test whether residency exposure adds to the medical school imprint. This appendix section documents the match procedure that links our Medicare cardiologists to Doximity profiles, maps Doximity training-program strings to AHA hospital identifiers, and constructs the residency-stage exposure measures used in Table ??.

E.1 Doximity Profile Data

Doximity (doximity.com) operates an online physician directory in which U.S. physicians can post professional profile pages. The profile pages report the physician's name, location, hospital affiliations, and credentials, along with structured fields for medical school, residency, and fellowship program names and graduation years. We work with the parsed profile fields that contain professional credentials, including the institution name and graduation year for each training stage, the cardiologist's current city and state, and gender. The profiles contain no patient-level information and no other identifiable personal data beyond the physician's professional credentials.

E.2 Cardiologist Linkage

The Doximity profiles do not include the National Provider Identifier, so we link cardiologists in our Medicare panel to Doximity profiles via normalized name, graduation year, and current practice state. We first normalize each side's name fields by transliterating to ASCII, lowercasing, stripping punctuation and credential suffixes (MD, DO, MPH, etc.), and collapsing whitespace. We then block on the normalized last and first name and apply a three-stage match.

Stage 1 retains pairs that are uniquely matched on normalized name from both sides. These represent the cleanest evidence of a real match. Stage 2 disambiguates pairs in which one cardiologist matches multiple Doximity profiles by name alone, using exact graduation-year agreement first, then state agreement, then exact medical-school string agreement. Stage 3 expands the candidate set for cardiologists not matched in Stage 1 or 2 using a small nickname canonicalization table (mapping common variants like Bob → Robert) and a first-initial fallback that requires exact graduation year and state agreement. Each downstream match must produce a one-to-one bipartite pairing.

Across all stages, the procedure matches 8,106 cardiologists from our Medicare panel, or roughly 75% of cardiologists with non-missing name and graduation year in Physician Compare. The Stage-1 unique-name matches form the validation set for the linkage. Stage

1 used only first and last name and did not condition on graduation year, state, gender, or medical school. Across this validation set, we observe 99.2% gender agreement, 86.5% graduation-year agreement (with most disagreements within one year), 34.5% practice-state agreement (consistent with Doximity profiles reflecting current practice in 2025–2026 while Physician Compare reflects practice locations in 2013–2018), and 29.1% exact medical-school string agreement (with most disagreements reflecting formatting differences between sources rather than substantive disagreement). The strong gender and graduation-year agreement on orthogonal fields supports the validity of the name-based block.

E.3 Doximity-to-AHA Hospital Mapping

We map each Doximity training-program string to an AHA hospital identifier in three tiers. First, a hand-curated override table covers approximately 130 distinct training-program strings, including all programs that account for more than approximately 25 cardiologists in our matched sample. Each override pairs the Doximity program name with a canonical AHA name fragment and the state in which the program operates, and the override is resolved by searching the AHA roster within that state for the matching name fragment. Second, we attempt exact matches on the normalized hospital name between Doximity and AHA. Third, we apply a substring fuzzy match that requires the Doximity name to contain or be contained by an AHA hospital name, with a length-ratio guardrail of 3.0 to reject implausible alignments such as a short Doximity name matching a much longer AHA name. The cardiologist-level coverage of the three-tier procedure is roughly 48% manual, 4% exact, 18% fuzzy, and 25% unmatched on the residency side, with similar coverage on the fellowship side. We report the main residency-stage results both for the full matched sample and for the cleaner manual-plus-exact subsample, which drops the fuzzy-tier matches to test sensitivity to the noisier matches.

E.4 Residency Exposure Measures

For each matched cardiologist, we construct two residency-stage exposure measures from the AHA Annual Survey. The own-hospital cath measure is the mean of the AHA cath lab indicator at the cardiologist's residency hospital over their first three years of post-graduate training, where the years are derived from the cardiologist's medical-school graduation year and a three-year residency window. The same-HRR-system share measure is the mean across the same window of the share of cath-equipped hospitals among all hospitals in the same AHA-coded health system as the residency hospital, restricted to hospitals within the same hospital referral region. For residency hospitals not affiliated with a multi-hospital system, or whose system has only one hospital in the same HRR, the same-HRR-system share is set equal to the own-hospital cath measure.

E.5 Direct-Exposure Test on Recent Graduates

The AHA cath lab indicator is a coarse proxy for procedural exposure during residency. A sharper measure is the residency hospital's realized NSTEMI catheterization rate during the trainee's first three years of post-graduate training, constructed directly from the claims data. We implement this measure for cardiologists with medical-school graduation year 2006 or later, whose first three post-graduate years (PGY1–PGY3) overlap the 2008–2018 claims window. The recent-graduate subsample includes 266 cardiologists and 487 cardiologist-years, with a mean of 2.05 PGY years observed per cardiologist. Table E.3 reports three specifications, progressing from practice-HRR and year fixed effects only (column 1) to the within-origin specification that adds medical-school HRR fixed effects (column 2) and a restriction to the cleaner manual-plus-exact residency-match subsample (column 3). The residency exposure coefficient is statistically indistinguishable from zero in all three columns. Point estimates are mildly negative but imprecisely estimated, and the confidence intervals do not support a positive residency imprint of the magnitude detected for medical school. The direct-exposure test on recent graduates corroborates the headline pipeline finding that residency exposure does not add to the medical school imprint.

Table E.3. Residency cath rate and own catheterization, recent-graduate subsample

	(1) Practice HRR FE	(2) Within origin	(3) Manual+exact only
Residency cath rate (PGY1-3)	-0.083 (0.087)	-0.197 (0.143)	-0.175 (0.150)
Cardiologist-years	442	254	223
Cardiologists	266	266	235
Practice HRR FE	Yes	Yes	Yes
Medical-school HRR FE	No	Yes	Yes
Year FE	Yes	Yes	Yes

F Selection Diagnostics

The main analysis addresses two distinct selection concerns. The first is that movers and stayers differ on observable characteristics, raising the possibility that the imprint estimate reflects mover-vs-stayer selection rather than a causal training effect. The second is that cardiologists from cath-rich training environments may systematically end up in cath-rich practice destinations, in which case our cross-sectional identification would conflate training imprint with destination-type sorting. We address the first with the inverse-probability-weighted (IPW) specification reported in the main paper. This section reports a parallel IPW exercise using a complementary measure of training-environment intensity, and a separate diagnostic for the destination-sorting concern.

F.1 IPW Robustness Using the Peer-Cath Measure

The main paper reports the IPW-weighted within-origin specification on the AHA training-period cath lab measure. Table F.4 reports a parallel IPW exercise using the leave-one-out peer cath rate in the medical school HRR as a complementary measure of training-environment intensity. The IPW weighting moves the imprint coefficient modestly relative to the unweighted within-origin baseline, in the same direction and of comparable magnitude to the shift we observe with the AHA measure in the main paper. Both exercises indicate that observable selection on graduation cohort, gender, subspecialty, and training-period intensity does not drive the imprint we estimate.

Table F.4. Selection robustness via inverse probability weighting (peer-cath measure)

	Baseline	IPW
Med school HRR intensity	0.076	0.087
	(0.060)	(0.059)
Δ intensity	0.030	0.025
	(0.028)	(0.028)
Practice HRR FE	Yes	Yes
Year FE	Yes	Yes
IPW	No	Yes
Observations	13,174	13,174

F.2 Training Intensity and Destination Choice

A separate identification concern is that cardiologists from cath-rich training environments may end up in cath-rich current peer environments. If so, the training imprint we estimate in the main paper could partly reflect destination-type sorting on the training measure rather than a causal effect of training. To assess this directly, we regress each cardiologist's current peer cath rate (the leave-one-out residualized cath rate of other cardiologists in the same HRR-year) on their training-period cath lab share, with year fixed effects and clustering at the medical school HRR level.

On the full cardiologist-year panel (10,736 observations from 3,207 cardiologists with non-missing training-period and current-peer data), the coefficient on training-period cath lab share is -0.012 (SE 0.016), statistically indistinguishable from zero. Restricting to each cardiologist's earliest panel observation as a cross-section produces a similarly small estimate: -0.027 (SE 0.018), $n = 3{,}207$. The pairwise correlation between training-period cath lab share and current peer cath rate is -0.03.

Training-period cath lab availability does not predict the current peer cath rate in our sample. Cardiologists from cath-rich training environments are not systematically located in cath-rich destinations, and the destination-type-sorting concern does not appear to drive our cross-sectional imprint estimate. We caution that our cardiologist-level cross-section uses the earliest panel observation rather than the first practice location after training, so this is a contemporaneous diagnostic rather than a direct test of training-induced sorting at first placement.

G Event-Study Details

This section reports the full coefficient table for the within-physician event study around mid-career moves.

Table G.5 reports point estimates, standard errors, and 95% confidence intervals for each event-time coefficient $\hat{\beta}_\tau$ in the within-physician event study, with $\tau \in [-3, 5]$ and $\tau = -1$ omitted as the reference period. The pre-move coefficients ($\tau \in \{-3, -2\}$) are statistically indistinguishable from zero, consistent with the parallel-trends assumption. Adaptation jumps at $\tau = 0$ to 0.443 (SE 0.097, 95% CI [0.25, 0.63]) and persists through $\tau = 5$. The $\tau \in \{2, 3\}$ coefficients are smaller and noisier than the $\tau = 0, 1, 4, 5$ coefficients, but the post-move trajectory is broadly stable rather than fading.

Pre-trends are formally tested by the joint F-statistic on $\hat{\beta}_{-3}$ and $\hat{\beta}_{-2}$, which fails to reject the null of zero pre-trends at conventional levels. The post-move pooled coefficient (constraining $\beta_\tau = \beta$ for $\tau \geq 0$) is 0.350, which we use in the cohort-transition calibration in the main paper.

Table G.5. Event-study coefficients, within-physician adaptation around mid-career moves

Event time τ	Estimate	SE	95% CI lo	95% CI hi	p-value
-3	0.063	0.136	-0.20	0.33	0.645
-2	-0.031	0.113	-0.25	0.19	0.785
-1	0				ref.
0	0.443	0.097	0.25	0.63	<0.001
1	0.312	0.114	0.09	0.54	0.006
2	0.184	0.129	-0.07	0.44	0.154
3	0.202	0.128	-0.05	0.45	0.116
4	0.375	0.139	0.10	0.65	0.007
5	0.376	0.163	0.06	0.69	0.021

H Cohort-Transition Simulation

The main paper combines $\hat{\beta}_{\text{train}} = 0.058$ and $\hat{\beta}_{\text{dest}} = 0.350$ into a steady-state multiplier of approximately 1.54. This section reports the derivation of the multiplier and the cohort-transition path implied by gradual cohort replacement.

H.1 Derivation of the Long-Run Multiplier

The multiplier in the main paper rests on three assumptions. First, the destination response operates linearly on the peer mean, so that a one-percentage-point shift in peer-environment cath rate raises own cath rate by β_{dest} percentage points. Second, the system reaches a steady state in which the peer mean equals the average own cath rate across all cardiologists in the HRR. Third, all cohorts contribute equally to the peer mean once the transition is complete. Let $\bar{y}$ denote the steady-state per-cardiologist cath rate and $\Delta\tau$ the uniform training-period cath lab shift. Each cardiologist's steady-state cath rate equals their direct training imprint plus their adaptation to the peer mean,

$$\bar{y} = \beta_{\text{train}} \cdot \Delta\tau + \beta_{\text{dest}} \cdot \bar{y},$$

which solves to

$$\bar{y} = \frac{\beta_{\text{train}} \cdot \Delta\tau}{1 - \beta_{\text{dest}}}.$$

The multiplier $1/(1 - \beta_{\text{dest}})$ captures the cumulative peer-effect spillover that accumulates as new cohorts displace old ones and incumbents adjust to the evolving peer environment.

H.2 Cohort Transition Path

We trace the per-cardiologist response to a uniform 10-percentage-point increase in training-period cath lab share for all newly entering cohorts, holding the cohort-replacement rate at 5% per year (corresponding to a 20-year career length). Each year, the share of cohorts that received the training shift rises by 5 percentage points until the entire cohort distribution has been replaced at year 20. The cardiologist-level cath rate evolves according to

$$y_t = \beta_{\text{train}} \cdot \Delta\tau \cdot \text{share}_t + \beta_{\text{dest}} \cdot y_{t-1},$$

with the direct training effect compounding through the destination feedback loop. The path approaches the long-run multiplier $1/(1 - \beta_{\text{dest}}) = 1.54$ asymptotically under our within-physician estimate of $\beta_{\text{dest}} = 0.35$. Roughly half of the long-run effect is realized within ten years of the shock, and the system reaches 95% of steady state by year 20.

References

Molitor, David, "The evolution of physician practice styles: evidence from cardiologist migration," *American Economic Journal: Economic Policy*, 2018, *10* (1), 326–56.